

# Who Pays More, and Who Can Least Afford It? Food Price Premiums, Deprivation and Abnormal Price Movements Across Selected Pakistani Urban Markets

**Milestone 02: Data preparation, EDA and hypotheses (draft Sections II and III of the manuscript)**

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Notebook, code and data: [github.com/Samzzh0803/Pakistan-Food-Price-Premiums-and-Deprivation](https://github.com/Samzzh0803/Pakistan-Food-Price-Premiums-and-Deprivation) (branch milestone-02)

**Summary.** Pakistan's weekly Sensitive Price Indicator (SPI) is reported as one national figure, but PBS also publishes prices for 51 items in 17 cities. We built a master dataset of 23,409 city-item-week rows from 27 weekly releases (2025-05-15 to 2026-09-24) and checked it against PBS's own national columns in all 1,377 item-weeks. We merged district deprivation (PSLM 2019-20), population (Census 2023) and urban food CPI. Across 32 food items, Islamabad pays +14.1% and Rawalpindi +10.2% relative to the national reference, while Sukkur (-7.1%) and Bannu (-7.4%) pay less. Perishables vary most across cities and branded goods least. City premiums are *negatively* rank-correlated with a four-indicator district deprivation composite (Spearman  $\rho = -0.56$ ,  $p = 0.020$ ,  $n = 17$ ), so the cities paying most are mostly not the ones least able to absorb it. On food insecurity alone the correlation is weaker and not significant ( $\rho = -0.32$ ,  $p = 0.209$ ), so this pattern is indicative, not established. We state five hypotheses for Milestones 03-04.

## 1. Project alignment and master dataset inventory

**Team.** Both members are computer scientists. One owns acquisition, parsing and provenance; the other owns statistical framing, representation, ethics and policy interpretation. **Research question.** Beneath the single national weekly SPI figure, do persistent city-level food price differences exist, are the cities paying most the ones least able to absorb it, and is the published city detail enough to detect localised price stress? **SDGs:** primary SDG 2 (Target 2.1, access to food; Target 2.c, functioning food markets and market information); secondary SDG 10 (reduced inequalities).

**Changes since Milestone 01.** The 7-city panel reported in Milestone 01 was a parsing error: Appendix-A stacks three tables in one sheet, and the old script read only the first. The panel covers all 17 SPI cities (8 Punjab, 4 Sindh, 2 KP, 2 Balochistan, plus Islamabad). A correction note is in the repository. Following the instructor's feedback, three external sources are now merged.

**Table 1. Master dataset: 23,409 rows × 46 columns. Observational unit: 1 row = 1 item's retail price in 1 city in 1 week (17 cities × 51 items × 27 weeks; 32 of the 51 items are food).**

| Source                                        | Role                                                                   | Coverage                           | Access                              |
|-----------------------------------------------|------------------------------------------------------------------------|------------------------------------|-------------------------------------|
| PBS weekly SPI Annexure, Appendix-A           | City-item prices (MIN/AVG/MAX): the panel                              | 27 weeks, 2025-05-15 to 2026-09-24 | xlsx from pbs.gov.pk release pages  |
| PSLM District Level Survey 2019-20            | District deprivation: FIES food insecurity, literacy, schooling, water | 2019-20                            | PDF report tables (text extraction) |
| Digital Census 2023, Table 1                  | Total and urban population                                             | 2023                               | xlsx, one file per province         |
| CPI (Urban), food and non-alcoholic beverages | Deflator only, not a regressor                                         | monthly, 2025-05 to 2026-09        | PDF and Monthly Review docx         |

## 2. Data ingestion, cleaning and structural readiness

### 2.1 Collection audit

Each PBS weekly release page (/weekly-sensitive-price-indicator-spi-for-the-week-ended-on-DD-MM-YYYY/) links an Annexure workbook. A rate-limited script tried the Thursday, Wednesday and Friday page for each of 73 weeks from 2025-05-15 and logged every request, URL and SHA-256 hash. It found release pages for 12 weeks, most of them weeks already retrieved for Milestone 01 (the re-downloads were byte-identical). After dropping 12 duplicate files, the panel holds 27 weeks. Other weeks may exist under URLs we did not try, so we do not treat the archive as complete. The week after our freeze date (2026-10-01) was stored, unopened, as a holdout. External files were downloaded from pbs.gov.pk and are kept unchanged with their URLs and hashes.

**Validation.** The parser asserts, for every week: 3 blocks, 17 cities, 51 items, 867 rows; no duplicate key;  $\text{MIN} \leq \text{AVG} \leq \text{MAX}$ ; and zeros only as all-zero triples. Against PBS's own national columns, the cross-city minimum and maximum match PBS's national MIN and MAX in 1,377 of 1,377 item-weeks. The **geometric mean** of the city averages matches PBS's national average within 0.1% in 1,377 of 1,377. The arithmetic mean misses by a median of 0.42%. We therefore use the geometric mean as the reference price.

### 2.2 Reshaping and alignment

The raw sheet (179 rows × 25 columns) is not tidy. City names are column headers, each spanning MIN/AVG/MAX cells, and three tables are stacked in one sheet. We **melt** each block into one row per city × item, **concatenate** the blocks and weeks, and move PBS's national columns into a separate validation table. We then **join** the item classification (on `item_id`), the city attributes (on `city`) and the CPI (on the month of `week_end`). Every join is many-to-one, and the row count is asserted unchanged (Table 2).

#### Key alignment decisions.

- **Karachi:** one SPI market, but six PSLM districts and seven census districts. PSLM values are population-weighted.
- **Islamabad and Rawalpindi:** one contiguous market, kept as two administrative units.

- **District totals:** PSLM has no urban sample for Khuzdar and publishes FIES only for district totals, so all indicators use district totals. These are **deprivation measures, not income**.

**Table 2. Reshaping and merge steps with keys, cardinality and the asserted row count.**

| operation | source                                  | key                        | cardinality                            | rows_after |
|-----------|-----------------------------------------|----------------------------|----------------------------------------|------------|
| stack     | 3 Appendix-A blocks per workbook        | same 51 item rows          | rows: 7+7+3 cities x 51 = 867 per week | 867        |
| stack     | 27 weekly workbooks                     | week_end                   | 867 x 27                               | 23409      |
| merge     | item_mapping.csv                        | item_id                    | many-to-one (51 items)                 | 23409      |
| merge     | city table (Census 2023 + PSLM 2019-20) | city                       | many-to-one (17 cities)                | 23409      |
| merge     | urban food CPI                          | calendar month of week_end | many-to-one (months)                   | 23409      |

## 2.3 Cleaning and missing data

Nothing is deleted. Zero MIN/AVG/MAX triples (83 rows, 0.35%) are set to missing and flagged: rice IRRI-6/9 is never quoted in Gujranwala, Lahore or Sialkot, and Khuzdar has no gas quote in two weeks. PBS itself excludes these cities from its national mean. Missing counts and percentages for **every column** are in Appendix Table A1. Most of the missingness is structural: weekly change columns are undefined across non-consecutive weeks (44.3% of rows), because only 15 pairs of weeks are 6-8 days apart. Table 3 gives the choice made for each kind of missingness. **Selection bias:** we exclude structural non-quotes from price comparisons, so results describe quoting markets and may still reflect selection bias, because a missing quote does not prove an item is unavailable in that city. The gaps in weekly coverage also mean the volatility analysis sees only the weeks we could retrieve.

**Table 3. Kinds of missingness, the choice made for each, and why.**

| kind                                       | rows   | choice                              | reason                                                                                                       |
|--------------------------------------------|--------|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Structural non-quote (PBS printed 0/0/0)   | 83     | set to missing, flag, do not impute | Imputing invents a market price PBS does not record; PBS itself excludes these cities from its national mean |
| Weekly change across a non-consecutive gap | 10,367 | leave missing                       | Interpolation would manufacture week-to-week changes nobody observed                                         |
| Weeks not retrieved from the PBS site      | –      | treat as a coverage gap             | No source to impute from; stated as a limitation, not filled                                                 |

## 2.4 Transformations, scaling and encoding

**Log transformation.** Prices are compared in logs because premiums are multiplicative: a 10% premium means the same thing for salt and for a 20 kg flour bag. Within an item, the median skewness of prices falls from 0.22 (raw) to 0.02 (log). Prices pooled across items in PKR are not a meaningful distribution (skewness 2.13), because they mix units. The main variable is  $\text{rel\_price} = \log(\text{city price} / \text{national geometric mean})$ , which sums to zero across cities in each item-week. District urban population is strongly right-skewed (skewness 2.30, driven by Karachi). Its log ( $\log\_pop\_urban$ ) has skewness -0.72. **Scaling.** The four deprivation indicators are min-max scaled to 0-1 (higher = more deprived) and averaged into  $\text{deprivation\_composite}$ . Prices are z-scored within each item and week ( $\text{price\_z\_item\_week}$ ). All constants come from the training weeks only. **Encoding.** Nominal strings become 0/1 dummy flags with fixed reference levels: Punjab for province, storable staple for item category (Table 4). The model-ready file `data/processed/m02/model_ready_food.parquet` (14,607 food rows × 44 columns) carries 7 dummy columns. Cities will enter H1 with sum-to-zero coding, so each coefficient is a premium relative to the average city.

**Table 4. Dummy encoding example: province and item\_category strings (left) become 0/1 flags (right). Reference levels: Punjab and storable\_staple (all zeros).**

| city      | province    | item_category    | prov_ICT | prov_Sindh | prov_KP | prov_Balochistan | cat_perishable | cat_branded_packaged | cat_prepared_food |
|-----------|-------------|------------------|----------|------------|---------|------------------|----------------|----------------------|-------------------|
| Islamabad | ICT         | perishable       | 1        | 0          | 0       | 0                | 1              | 0                    | 0                 |
| Lahore    | Punjab      | perishable       | 0        | 0          | 0       | 0                | 1              | 0                    | 0                 |
| Karachi   | Sindh       | perishable       | 0        | 1          | 0       | 0                | 1              | 0                    | 0                 |
| Peshawar  | KP          | perishable       | 0        | 0          | 1       | 0                | 1              | 0                    | 0                 |
| Quetta    | Balochistan | perishable       | 0        | 0          | 0       | 1                | 1              | 0                    | 0                 |
| Islamabad | ICT         | branded_packaged | 1        | 0          | 0       | 0                | 0              | 1                    | 0                 |
| Islamabad | ICT         | prepared_food    | 1        | 0          | 0       | 0                | 0              | 0                    | 1                 |

## 3. Exploratory data analysis

### 3.1 Univariate summary

Table 5 reports each variable at its own unit of analysis. City attributes take 17 values, so city-level relationships are always analysed with  $n = 17$ , never with the 23,409 rows. **Skewed distributions** show up where the resistant and non-resistant measures disagree. The within-city price range has mean 6.8% but median 1.4% (skewness 3.55). District urban population has mean 3.52 million but median 1.62 million. For city premiums, Larkana's mean (-4.9%) and median (-0.2%) differ by 4.8 points, because a few items drive the mean. We report medians alongside means wherever the two disagree. Weekly price changes are dominated by zeros: 75% of consecutive-week food changes are exactly 0, so their median and IQR are both 0.

**Table 5. Summary statistics for key numerical variables: central tendency (mean, median), spread (SD, IQR) and skewness.**

| variable (unit of analysis; unit)                | n     | mean   | median | sd     | iqr    | skew   | min    | max     |
|--------------------------------------------------|-------|--------|--------|--------|--------|--------|--------|---------|
| price_z_item_week (food; z-score, unitless)      | 14539 | -0.000 | -0.043 | 0.970  | 1.227  | 0.189  | -3.865 | 3.795   |
| rel_price (food; log points)                     | 14607 | -0.000 | 0.000  | 0.130  | 0.110  | -0.144 | -1.239 | 0.708   |
| dlog_price (food; log points, consecutive weeks) | 8115  | -0.003 | 0.000  | 0.072  | 0.000  | -1.300 | -0.981 | 0.709   |
| within_city_range_pct (food; %)                  | 14607 | 6.812  | 1.395  | 11.665 | 9.384  | 3.549  | 0.000  | 152.858 |
| single_quote (food; 1 if MIN = MAX, else 0)      | 14607 | 0.486  | 0.000  | 0.500  | 1.000  | 0.056  | 0.000  | 1.000   |
| fies_mod_sev_pct (17 cities; %)                  | 17    | 15.314 | 16.380 | 5.926  | 8.310  | 0.647  | 7.930  | 29.260  |
| illiteracy10_pct (17 cities; %)                  | 17    | 35.756 | 34.000 | 15.262 | 20.155 | 1.080  | 15.000 | 78.000  |
| out_of_school_pct (17 cities; %)                 | 17    | 28.573 | 25.739 | 15.495 | 19.000 | 0.627  | 10.000 | 60.000  |
| no_tap_water_pct (17 cities; %)                  | 17    | 76.782 | 81.740 | 20.667 | 18.130 | -1.121 | 31.680 | 100.000 |
| deprivation_composite (17 cities; 0-1)           | 17    | 0.427  | 0.349  | 0.217  | 0.261  | 0.840  | 0.138  | 0.974   |
| pop_urban (17 city districts; millions)          | 17    | 3.524  | 1.619  | 4.940  | 2.485  | 2.298  | 0.048  | 18.868  |
| log_pop_urban (17 city districts; log persons)   | 17    | 14.368 | 14.298 | 1.341  | 1.176  | -0.725 | 10.787 | 16.753  |

### 3.2 City food price premiums

For each city we average `rel_price` over weeks for each food item. The item-level means are the observations, so `n` is 31-32 items, not thousands of rows. Islamabad's mean premium is +14.1% (95% t-interval [8.9, 19.6]), followed by Rawalpindi +10.2% and Karachi +5.2%. Sukkur (-7.1%) and Bannu (-7.4%) are lowest (Fig. 1). 8 of 17 cities differ from zero at  $\alpha = 0.05$ . After a Bonferroni correction for 17 tests, 4 remain: Islamabad, Rawalpindi, Sukkur, Bannu. Premiums persist: of 3,269 consecutive-week pairs in which a city was above the cross-city median, 92.7% stayed above it the next week. Restricting to pairs where the price actually changed, the figure is 84.1%.

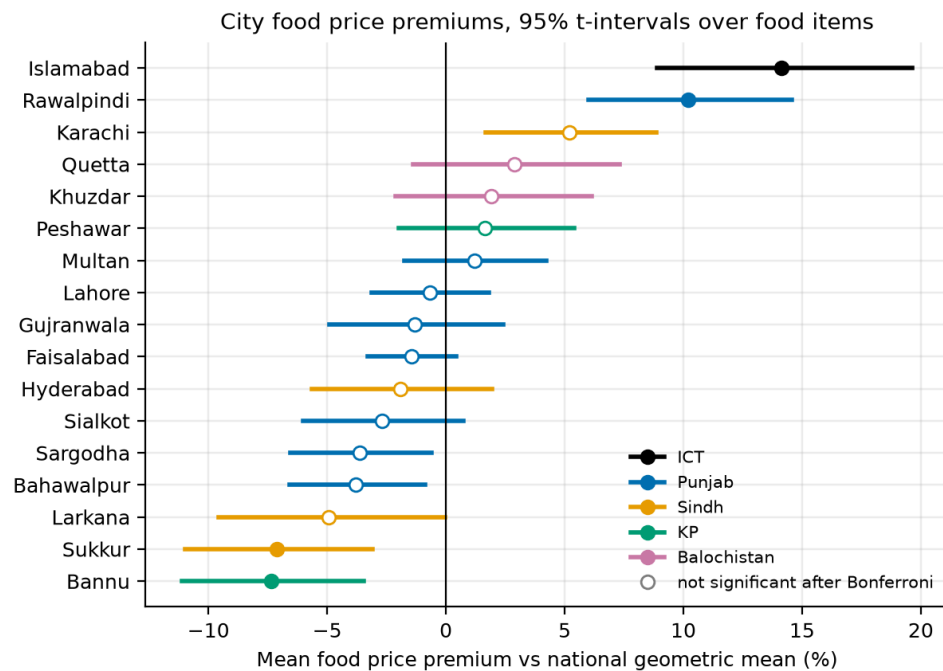


Fig. 1. Mean food price premium of each city relative to the national geometric mean, with 95% t-intervals over food items ( $n = 31-32$ ). Colour shows province; filled markers remain significant after Bonferroni correction ( $\alpha = 0.05/17$ ). Islamabad and Rawalpindi pay most; Sukkur and Bannu least.

### 3.3 Dispersion by food category

The coefficient of variation (CV) puts items priced in different units on one scale. Mean cross-city CV is highest for perishables (0.157, 11 items) and prepared food (0.144), lower for storable staples (0.107, 10 items), and lowest for branded packaged goods (0.026), which carry printed or distributor prices (Fig. 2). A one-tailed Welch comparison of perishables against storables gives  $p = 0.037$ , and a rank-based check gives  $p = 0.057$ . With about ten items per group this is suggestive, not conclusive. Categories were assigned on storability and pricing mechanism before any dispersion was computed.

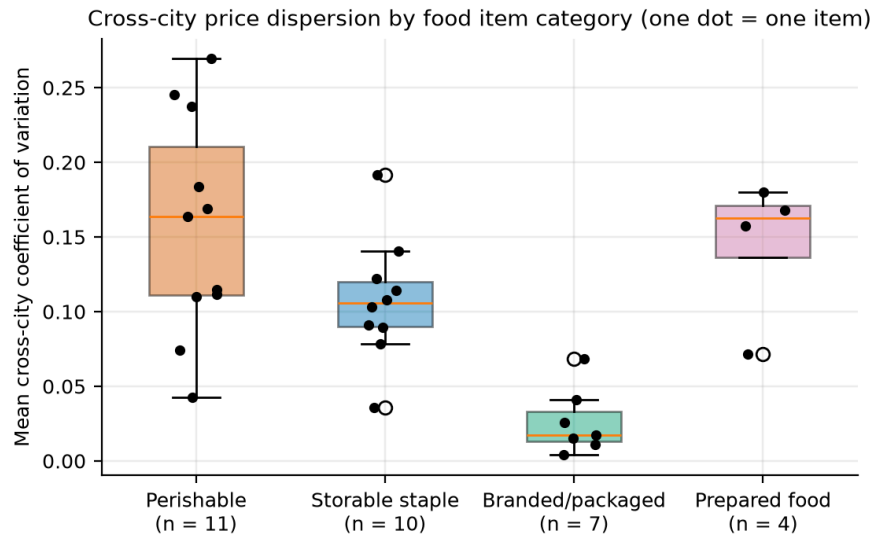


Fig. 2. Mean cross-city coefficient of variation (unitless) of each food item, grouped by category; one dot per item. Perishables are the most dispersed across cities; branded packaged goods barely differ.

### 3.4 Outlier diagnostic

We apply both rules taught in the course to `rel_price` and to weekly log changes, each at two tiers (Table 6, Fig. 3). **Masking and variance inflation:** the two rules disagree sharply. On `rel_price`, the IQR is narrow because many branded items are priced identically everywhere, so the quartile fences sit close and flag 3,649 values. The SD, by contrast, is inflated by the extreme values themselves, which pulls the 3-SD fence outward and masks them: only 434 are flagged. On weekly changes the IQR is exactly 0, so the quartile rule flags every non-zero change and is unusable. We checked the eight most extreme relative prices against the raw workbooks. 7 equal PBS's own national minimum or maximum for that item-week, and the other is a genuine printed quote (Sukkur tomatoes, PKR 20 per 1 kg). They are therefore true prices: we keep and flag them rather than trim them, because large moves are what H3 studies.

Table 6. Outliers by the quartile rule (beyond  $Q1 - k \cdot IQR$  or  $Q3 + k \cdot IQR$ ) and the z-score rule ( $|z| > 2, 3$ ). Units: log points.

| variable                     | n     | Q1     | Q3    | IQR   | mean  | sd    | > 1.5 IQR | > 3 IQR | $ z  > 2$ | $ z  > 3$ |
|------------------------------|-------|--------|-------|-------|-------|-------|-----------|---------|-----------|-----------|
| rel_price, all 51 items      | 23326 | -0.037 | 0.046 | 0.084 | 0.000 | 0.133 | 3649      | 1251    | 1598      | 434       |
| weekly log change, all items | 12959 | 0.000  | 0.000 | 0.000 | 0.001 | 0.062 | 2502      | 2502    | 493       | 296       |

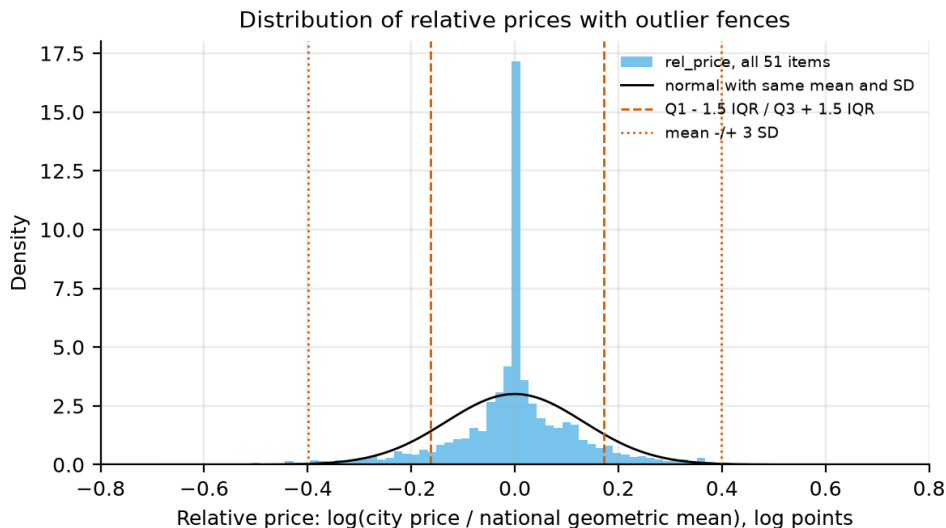


Fig. 3. Distribution of relative prices (log points; all 51 items  $\times$  17 cities  $\times$  27 weeks) with the 1.5 IQR fences (dashed) and the  $\pm 3$  SD fences (dotted), plus a normal curve with the same mean and SD. The spike at 0 and the heavy tails explain why the two rules disagree.

### 3.5 Relationships: premium, deprivation and data quality

Following Anscombe's advice, we plot before computing any coefficient (Fig. 4). The relationship runs **against** the "poor pay more" expectation. The three least deprived cities (Karachi, Islamabad, Rawalpindi) have the highest premiums, while several of the most deprived (Bannu, Sukkur, Larkana) pay below the national reference. The Spearman correlation between premium and the deprivation composite is  $\rho = -0.56$  ( $p = 0.020$ ,  $n = 17$ ). Dropping each city in turn keeps  $\rho$  between -0.72 and -0.48. Against FIES food insecurity alone,  $\rho = -0.32$  ( $p = 0.209$ ), which is not significant. The tap-water indicator is a weak deprivation measure: it rates Gujranwala and Sialkot (94% and 95% of households without tap water) as worse off than Quetta or Peshawar, and Karachi and Hyderabad as best off. This mostly reflects reliance on motor pumps and filtration plants in Punjab

cities. Without it, the composite gives  $\rho = -0.50$  ( $p = 0.041$ ). Khuzdar and Quetta are the exception: their premiums are above zero, though not individually significant, and they have the highest food insecurity in the panel, so burden concentrates there. These are associations across 17 cities, with deprivation data from 2019-20, and support no causal claim.

City food price premium against district deprivation (PSLM 2019-20)

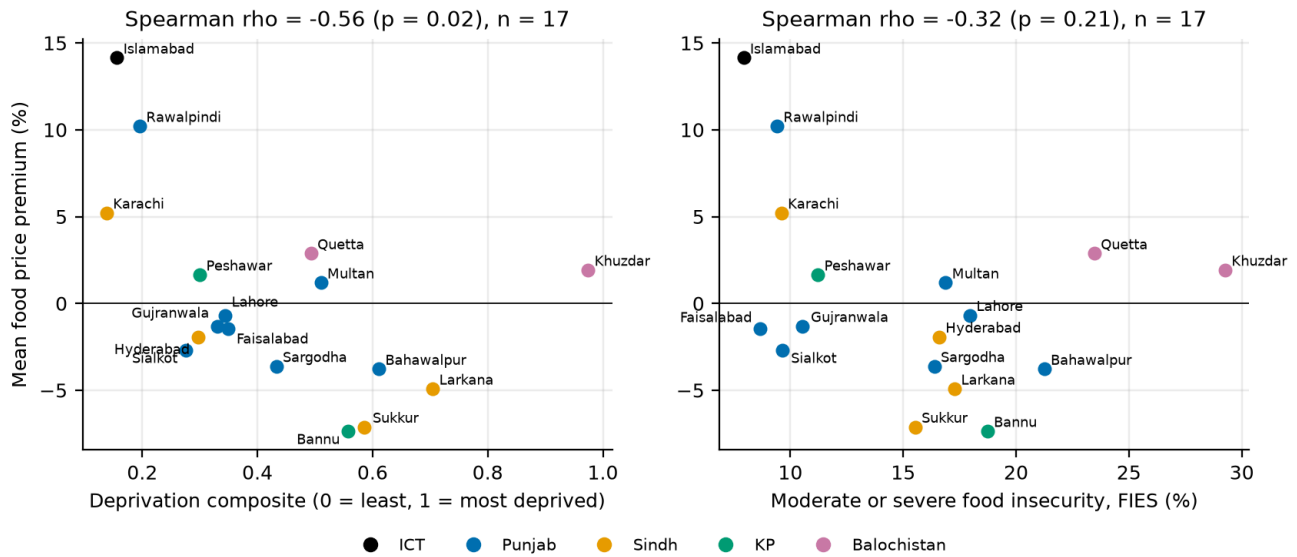


Fig. 4. City mean food price premium (%) against the PSLM 2019-20 deprivation composite (left; 0-1, higher = more deprived) and moderate-or-severe food insecurity (right; % of population). Every city is labelled; colour shows province;  $n = 17$ .

Table 7. Spearman correlation matrix of city-level variables ( $n = 17$ ).

| Spearman $\rho$<br>( $n = 17$ ) | premium | deprivation<br>composite | FIES food<br>insecurity | log urban<br>population | share<br>MIN =<br>MAX |
|---------------------------------|---------|--------------------------|-------------------------|-------------------------|-----------------------|
| premium                         | 1.00    | -0.56                    | -0.32                   | 0.41                    | -0.74                 |
| deprivation<br>composite        | -0.56   | 1.00                     | 0.76                    | -0.56                   | 0.49                  |
| FIES food<br>insecurity         | -0.32   | 0.76                     | 1.00                    | -0.41                   | 0.14                  |
| log urban<br>population         | 0.41    | -0.56                    | -0.41                   | 1.00                    | -0.41                 |
| share MIN =<br>MAX              | -0.74   | 0.49                     | 0.14                    | -0.41                   | 1.00                  |

Table 8. Contingency table: SPI cities by province and premium tercile.

| province    | low premium | mid premium | high premium |
|-------------|-------------|-------------|--------------|
| ICT         | 0           | 0           | 1            |
| Punjab      | 3           | 4           | 1            |
| Sindh       | 2           | 1           | 1            |
| KP          | 1           | 0           | 1            |
| Balochistan | 0           | 0           | 2            |

**Data quality.** Table 7 also shows that the premium correlates negatively with the share of a city's food records where MIN = MAX ( $\rho = -0.74$ ). MIN = MAX means every quote PBS collected for that item in that city was identical: either a single quote or several equal ones. The published average then carries no information about price spread within the city. This share ranges from 20% in Karachi to 91% in Sukkur (Fig. 5), a measurement-quality disparity that H4 carries forward. **Representation.** The census districts that contain the 17 SPI cities hold 65% of the urban population of the four provinces and Islamabad. Because each city is smaller than its district, this is an upper bound on what the SPI markets represent. The panel includes no rural markets, AJK or Gilgit-Baltistan, and only two cities each for KP and Balochistan.

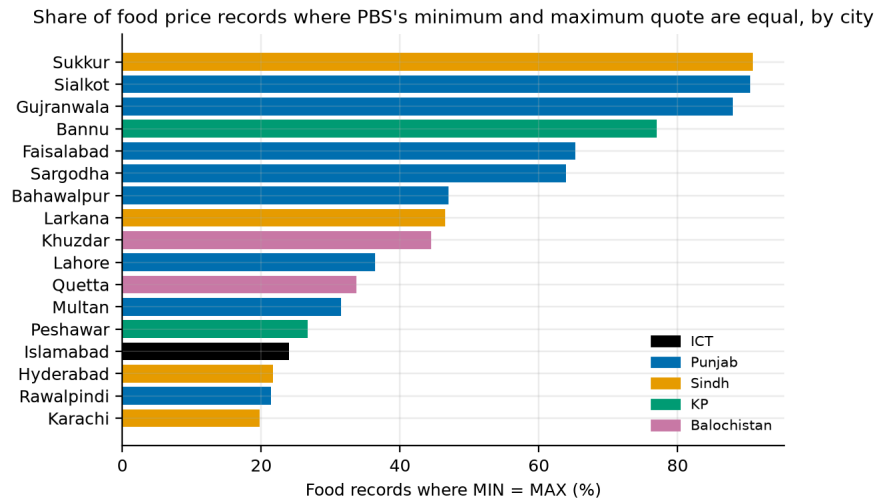


Fig. 5. Share of each city's food price records (%) in which PBS's minimum and maximum quote are equal, coloured by province. High shares mean the published city average carries no information on within-city price spread.

#### 4. Hypotheses and analytical plan

Each hypothesis is mapped to master-dataset columns in Table 9. Targets marked † are derived from existing columns, and the rule for each is fixed now. `abnormal_next_week` has been derived provisionally in the notebook, which is the source of the base rate and row count quoted below, using thresholds fitted on training weeks only. The final label and the model-error columns will be built in Milestone 03.

- **H1, persistent premiums (SDG 2.1).** (a)  $\ln \log p_{c,i,t} = \alpha_{i,t} + \beta_c + \varepsilon$  with  $\Sigma \beta_c = 0$ : H0:  $\beta_c = 0$  for every city  $c$ ; H1:  $\beta_c \neq 0$  for at least one  $c$ . (b) H0:  $P(\text{above median at } t+1 \mid \text{above at } t) = 0.5$ ; H1:  $> 0.5$ , tested on consecutive weeks and repeated for changed prices only. If some cities persistently pay more for the same basket, access to food differs by place in a way the national figure hides.
- **H2, perishability and dispersion (SDG 2.c).** H0:  $\mu_{CV, \text{perishable}} = \mu_{CV, \text{storable}}$ ; H1:  $\mu_{CV, \text{perishable}} > \mu_{CV, \text{storable}}$ . Greater dispersion in perishables points to market-functioning frictions.
- **H3, abnormal next-week movement (SDG 2.c, early warning).** H0: a classifier on lagged features achieves AUC = 0.5 on a time-ordered test set; H1: AUC > 0.5. "Abnormal" means a change larger than the item's own 90th percentile of non-zero changes in the training weeks. We use non-zero changes because 75% of weekly food price changes are zero, and the plain 90th percentile would be 0 for 13 of the 32 food items. Base rate: 2.9%.
- **H4, representation and model fairness (SDG 10).** H0: model error rates and data quality (share of MIN = MAX records) are equal across city groups (province, population tercile, deprivation tercile); H1: at least one group differs.
- **H5, burden versus price (SDG 2.1, SDG 10).** H0: Spearman  $\rho(\text{city premium, deprivation}) = 0$ ; H1:  $\rho \neq 0$ , two-sided,  $n = 17$ . EDA already points to  $\rho < 0$  (§3.5). The test will be repeated under both deprivation definitions.

Table 9. Variable mapping: target, predictors and controls for each hypothesis, with column names from the master dataset or its views.

| Hypothesis        | Target (Y)                                                        | Predictors (X)                                                                                                                                                                                             | Controls / confounders                                                                          | Data                                   | Effective n                    |
|-------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------|
| H1a premiums      | <code>log_price</code>                                            | <code>city</code> (sum-to-zero dummies of <code>city_code</code> )                                                                                                                                         | <code>item × week</code> fixed effects ( <code>item_id</code> , <code>week_end</code> )         | master, food rows                      | 17 cities; 32 items × 27 weeks |
| H1b persistence   | <code>above_median</code>                                         | <code>above_median_prev</code>                                                                                                                                                                             | <code>price_changed</code> , <code>item_category</code> ; rows with <code>is_consecutive</code> | master, food rows                      | 17 cities (one rate each)      |
| H2 dispersion     | <code>cv</code> , averaged per item                               | <code>item_category</code> (perishable vs storable_staple)                                                                                                                                                 | <code>n_quoting</code>                                                                          | <code>item_week_dispersion</code> view | 11 vs 10 items                 |
| H3 abnormal move  | <code>abnormal_next_week</code> † (to be derived in M03)          | <code>dlog_price</code> , <code>dlog_price_lag1</code> , <code>price_changed</code> , <code>rel_price</code> , <code>within_city_range_pct</code> , <code>single_quote</code> , <code>cat_*</code> dummies | <code>prov_*</code> dummies, <code>log_pop_urban</code> , <code>iso_week</code>                 | <code>model_ready_food</code>          | 7,095 labelled rows            |
| H4 representation | model error † (M03); <code>single_quote</code> (now)              | province; terciles of <code>pop_urban</code> and <code>deprivation_composite</code> †                                                                                                                      | <code>item_category</code>                                                                      | master / city table                    | 17 cities in 3-5 groups        |
| H5 burden         | <code>city premium</code> = city mean of <code>rel_price</code> † | <code>deprivation_composite</code> , <code>fies_mod_sev_pct</code>                                                                                                                                         | <code>log_pop_urban</code> , city share of <code>single_quote</code>                            | city table (17 rows)                   | 17                             |

**Analytical plan.** Within the course's agreed scope (classification, regression, clustering and fairness analysis), Milestone 03 will:

- estimate H1 with a fixed-effects regression;
- fit logistic regression and tree-ensemble classifiers for H3;
- validate on a time-ordered split, because random folds would leak later weeks of the same city-item series;
- open the frozen holdout once, at the end;

- evaluate with AUC and precision-recall rather than accuracy, since a model that always predicts "normal" would be about 97% accurate;
- break all results down by city group for H4.

The notebook also contains exploratory clustering, PCA and nearest-neighbour analyses, which are not part of this report.

## 5. Limitations

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- **Coverage:** 27 weeks with gaps and only 15 consecutive pairs, which limits volatility analysis and the sample size for H3.
- **Sample:** 17 cities chosen by PBS, urban only. Results do not generalise to rural or small-town markets.
- **Deprivation:** 2019-20 district totals that include rural areas (Bannu's district is 3.6% urban). These are measures of deprivation, not city income.
- **Measurement:** SPI records retail prices only. PSLM values were extracted from PDF tables; each is stored with its page number for checking.

## References

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Pakistan Bureau of Statistics. Weekly Sensitive Price Indicator, Annexure workbooks, weeks ended 2025-05-15 to 2026-09-24. <https://www.pbs.gov.pk/>

Pakistan Bureau of Statistics. Pakistan Social and Living Standards Measurement Survey 2019-20, District Level: Tables 2.14(a), 2.15, 7.1, 9.1.

Pakistan Bureau of Statistics. 7th Population and Housing Census 2023 (Digital Census): Table 1, district results.

Pakistan Bureau of Statistics. CPI (Urban) Group-wise Cumulative Indices, base 2015-16; Monthly Review on Price Indices, August and September 2026.

Wickham, H. (2014). Tidy Data. *Journal of Statistical Software*, 59(10).

Anscombe, F. J. (1973). Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17-21.

## Appendix

**Table A1. Missing values (count and % of 23,409 rows) for every column of the master dataset, with Unit 02 attribute type. Weekly-change columns are missing wherever the previous week is not 6-8 days earlier. Price columns are missing only for the structural non-quotes. price\_z\_item\_week is undefined where every quoting city charges the same price (SD = 0). days\_since\_prev is missing in the first week.**

| column                | attribute type   | missing (n) | missing (%) | column                        | attribute type        | missing (n) | missing (%) |
|-----------------------|------------------|-------------|-------------|-------------------------------|-----------------------|-------------|-------------|
| week_end              | interval         | 0           | 0.00        | within_city_range_pct         | ratio                 | 83          | 0.35        |
| year                  | interval         | 0           | 0.00        | single_quote                  | binary                | 83          | 0.35        |
| month                 | ordinal (cyclic) | 0           | 0.00        | price_z_item_week             | interval              | 2,903       | 12.40       |
| iso_week              | ordinal (cyclic) | 0           | 0.00        | days_since_prev               | ratio                 | 867         | 3.70        |
| city                  | nominal          | 0           | 0.00        | is_consecutive                | binary                | 0           | 0.00        |
| city_code             | nominal          | 0           | 0.00        | dlog_price                    | interval              | 10,450      | 44.64       |
| province              | nominal          | 0           | 0.00        | price_changed                 | binary                | 10,450      | 44.64       |
| item_id               | nominal          | 0           | 0.00        | dlog_price_lag1               | interval              | 15,634      | 66.79       |
| item_label            | nominal          | 0           | 0.00        | pop_total                     | ratio                 | 0           | 0.00        |
| item_short            | nominal          | 0           | 0.00        | pop_urban                     | ratio                 | 0           | 0.00        |
| unit                  | nominal          | 0           | 0.00        | log_pop_urban                 | interval              | 0           | 0.00        |
| is_food               | binary (nominal) | 0           | 0.00        | fies_mod_sev_pct              | ratio                 | 0           | 0.00        |
| item_category         | nominal          | 0           | 0.00        | illiteracy10_pct              | ratio                 | 0           | 0.00        |
| price_min             | ratio            | 83          | 0.35        | out_of_school_pct             | ratio                 | 0           | 0.00        |
| price_avg             | ratio            | 83          | 0.35        | no_tap_water_pct              | ratio                 | 0           | 0.00        |
| price_max             | ratio            | 83          | 0.35        | deprivation_composite         | interval (0-1 scaled) | 0           | 0.00        |
| is_structural_missing | binary           | 0           | 0.00        | deprivation_composite_nowater | interval (0-1 scaled) | 0           | 0.00        |
| log_price             | interval         | 83          | 0.35        | cpi_urban_food                | ratio (index)         | 0           | 0.00        |
| national_gm           | ratio            | 0           | 0.00        | real_price_avg                | ratio                 | 83          | 0.35        |
| rel_price             | interval         | 83          | 0.35        | block                         | nominal               | 0           | 0.00        |
| cross_city_median     | ratio            | 0           | 0.00        | source_file                   | nominal               | 0           | 0.00        |
| above_median          | binary           | 83          | 0.35        | sha256                        | nominal               | 0           | 0.00        |
| above_median_prev     | binary           | 10,450      | 44.64       | parser_version                | nominal               | 0           | 0.00        |

**Table A2. City attributes: Census 2023 district urban population and PSLM 2019-20 district deprivation indicators (higher = more deprived).**

| city       | province    | district urban pop. (m) | FIES mod/sev % | illiterate 10+ % | out of school 5-16 % | no tap water % | composite (0-1) |
|------------|-------------|-------------------------|----------------|------------------|----------------------|----------------|-----------------|
| Islamabad  | ICT         | 1.11                    | 7.93           | 15.00            | 10.00                | 74.31          | 0.16            |
| Rawalpindi | Punjab      | 4.21                    | 9.41           | 18.00            | 10.00                | 77.16          | 0.20            |
| Gujranwala | Punjab      | 3.59                    | 10.53          | 26.00            | 15.00                | 95.06          | 0.33            |
| Sialkot    | Punjab      | 1.48                    | 9.65           | 21.00            | 11.00                | 93.78          | 0.28            |
| Lahore     | Punjab      | 13.00                   | 17.96          | 23.00            | 16.00                | 76.71          | 0.34            |
| Faisalabad | Punjab      | 4.39                    | 8.67           | 33.00            | 21.00                | 90.11          | 0.35            |
| Sargodha   | Punjab      | 1.61                    | 16.38          | 33.00            | 18.00                | 92.44          | 0.43            |
| Multan     | Punjab      | 2.50                    | 16.88          | 39.00            | 30.00                | 89.09          | 0.51            |
| Bahawalpur | Punjab      | 1.62                    | 21.25          | 51.00            | 32.00                | 86.87          | 0.61            |
| Karachi    | Sindh       | 18.87                   | 9.60           | 23.85            | 25.74                | 32.94          | 0.14            |
| Hyderabad  | Sindh       | 2.02                    | 16.59          | 34.00            | 34.00                | 31.68          | 0.30            |
| Sukkur     | Sindh       | 0.81                    | 15.53          | 46.00            | 48.00                | 81.74          | 0.59            |
| Larkana    | Sindh       | 0.80                    | 17.28          | 44.00            | 56.00                | 100.00         | 0.70            |
| Peshawar   | KP          | 1.91                    | 11.22          | 40.00            | 25.00                | 55.51          | 0.30            |
| Bannu      | KP          | 0.05                    | 18.73          | 47.00            | 39.00                | 75.19          | 0.56            |
| Quetta     | Balochistan | 1.57                    | 23.47          | 36.00            | 35.00                | 59.72          | 0.49            |
| Khuzdar    | Balochistan | 0.36                    | 29.26          | 78.00            | 60.00                | 92.98          | 0.97            |

**Table A3. Per-item summary over the 27 training weeks. Mean price = average of the quoting cities' average prices in each week, then averaged over weeks, in PKR per the PBS unit shown (e.g. eggs per dozen, wheat flour per 20 kg bag). Mean CV = cross-city coefficient of variation, averaged over weeks. Share unchanged = share of consecutive-week price changes that are exactly 0. Share MIN = MAX = share of city-week records where all quotes were equal.**

| PBS # | item                             | PBS unit   | category             | mean price (PKR per PBS unit) | mean CV | share unchanged wk/wk | share MIN = MAX |
|-------|----------------------------------|------------|----------------------|-------------------------------|---------|-----------------------|-----------------|
| 1     | Wheat flour 20kg                 | 20 Kg      | Storable staple      | 2,213.646                     | 0.108   | 0.494                 | 0.301           |
| 2     | Rice basmati broken              | 1 Kg       | Storable staple      | 218.454                       | 0.192   | 0.949                 | 0.240           |
| 3     | Rice IRRI-6/9                    | 1 Kg       | Storable staple      | 156.874                       | 0.141   | 0.943                 | 0.177           |
| 4     | Bread plain                      | Each       | Prepared food        | 113.288                       | 0.072   | 0.988                 | 0.919           |
| 5     | Beef with bone                   | 1 Kg       | Perishable           | 1,195.088                     | 0.110   | 0.906                 | 0.340           |
| 6     | Mutton                           | 1 Kg       | Perishable           | 2,211.312                     | 0.115   | 0.906                 | 0.344           |
| 7     | Chicken broiler (live)           | 1 Kg       | Perishable           | 379.066                       | 0.074   | 0.075                 | 0.549           |
| 8     | Milk fresh                       | 1 Ltr      | Perishable           | 208.180                       | 0.112   | 0.945                 | 0.442           |
| 9     | Curd (dahi) loose                | 1 Kg       | Perishable           | 245.620                       | 0.164   | 0.945                 | 0.458           |
| 10    | Powdered milk NIDO 390g          | Each       | Branded/packaged     | 1,128.550                     | 0.026   | 0.898                 | 0.706           |
| 11    | Eggs (farm)                      | 1 Dozen    | Perishable           | 289.193                       | 0.042   | 0.239                 | 0.458           |
| 12    | Mustard oil (avg quality)        | 1 Kg       | Storable staple      | 550.066                       | 0.122   | 0.894                 | 0.423           |
| 13    | Cooking oil 5L tin (brand)       | Each       | Branded/packaged     | 2,981.018                     | 0.015   | 0.875                 | 0.935           |
| 14    | Vegetable ghee 2.5kg tin (brand) | Each       | Branded/packaged     | 1,514.523                     | 0.017   | 0.894                 | 0.946           |
| 15    | Vegetable ghee 1kg pouch (brand) | Each       | Branded/packaged     | 589.763                       | 0.011   | 0.867                 | 0.935           |
| 16    | Bananas                          | 1 Dozen    | Perishable           | 170.233                       | 0.269   | 0.580                 | 0.275           |
| 17    | Pulse masoor                     | 1 Kg       | Storable staple      | 277.183                       | 0.090   | 0.788                 | 0.303           |
| 18    | Pulse moong                      | 1 Kg       | Storable staple      | 382.797                       | 0.091   | 0.769                 | 0.298           |
| 19    | Pulse mash                       | 1 Kg       | Storable staple      | 455.341                       | 0.079   | 0.792                 | 0.312           |
| 20    | Pulse gram                       | 1 Kg       | Storable staple      | 274.550                       | 0.103   | 0.678                 | 0.333           |
| 21    | Potatoes                         | 1 Kg       | Perishable           | 62.326                        | 0.237   | 0.537                 | 0.264           |
| 22    | Onions                           | 1 Kg       | Perishable           | 108.545                       | 0.184   | 0.259                 | 0.301           |
| 23    | Tomatoes                         | 1 Kg       | Perishable           | 138.434                       | 0.245   | 0.125                 | 0.327           |
| 24    | Sugar refined                    | 1 Kg       | Storable staple      | 165.730                       | 0.036   | 0.392                 | 0.429           |
| 25    | Gur                              | 1 Kg       | Storable staple      | 235.110                       | 0.114   | 0.843                 | 0.290           |
| 26    | Salt powdered 800g (brand)       | Each       | Branded/packaged     | 67.129                        | 0.068   | 0.965                 | 0.824           |
| 27    | Chilli powder 200g (brand)       | Each       | Branded/packaged     | 346.754                       | 0.041   | 0.965                 | 1.000           |
| 28    | Garlic                           | 1 Kg       | Perishable           | 421.912                       | 0.169   | 0.604                 | 0.233           |
| 29    | Tea Lipton 190g                  | Each       | Branded/packaged     | 450.141                       | 0.004   | 0.992                 | 0.904           |
| 30    | Cooked beef (hotel plate)        | Per Plate  | Prepared food        | 321.057                       | 0.168   | 0.984                 | 0.355           |
| 31    | Cooked daal (hotel plate)        | Per Plate  | Prepared food        | 172.415                       | 0.180   | 0.988                 | 0.327           |
| 32    | Tea prepared (cup)               | Per Cup    | Prepared food        | 64.085                        | 0.158   | 0.976                 | 0.556           |
| 33    | Cigarettes Capstan 20s           | Each       | Other non-food       | 246.273                       | 0.027   | 0.973                 | 0.797           |
| 34    | Long cloth 57in (brand)          | 1 mtr      | Other non-food       | 679.400                       | 0.163   | 0.996                 | 0.523           |
| 35    | Shirting                         | 1 mtr      | Other non-food       | 546.212                       | 0.293   | 0.973                 | 0.214           |
| 36    | Lawn printed (brand)             | 1 mtr      | Other non-food       | 676.896                       | 0.156   | 0.988                 | 0.458           |
| 37    | Georgette                        | 1 mtr      | Other non-food       | 316.572                       | 0.208   | 0.980                 | 0.370           |
| 38    | Gents sandal Bata                | Pair       | Other non-food       | 2,499.000                     | 0.000   | 1.000                 | 1.000           |
| 39    | Gents sponge chappal Bata        | Pair       | Other non-food       | 628.630                       | 0.000   | 1.000                 | 1.000           |
| 40    | Ladies sandal Bata               | Pair       | Other non-food       | 1,428.630                     | 0.000   | 1.000                 | 1.000           |
| 41    | Electricity Q1                   | Per Unit   | Administered/utility | 6.086                         | 0.000   | 0.733                 | 1.000           |
| 42    | Gas Q1                           | MMBTU      | Administered/utility | 2,522.796                     | 0.000   | 0.937                 | 1.000           |
| 43    | Firewood                         | 40 Kg      | Other non-food       | 1,471.809                     | 0.214   | 0.910                 | 0.298           |
| 44    | Energy saver bulb (brand)        | Each       | Other non-food       | 413.412                       | 0.290   | 0.976                 | 0.804           |
| 45    | Washing soap Sufi (brand)        | Each       | Other non-food       | 135.581                       | 0.046   | 0.988                 | 0.832           |
| 46    | Match box                        | Each       | Other non-food       | 6.297                         | 0.117   | 1.000                 | 0.824           |
| 47    | Petrol super                     | Per Litre  | Administered/utility | 305.818                       | 0.003   | 0.667                 | 0.745           |
| 48    | Hi-speed diesel                  | Per Litre  | Administered/utility | 326.530                       | 0.002   | 0.600                 | 0.747           |
| 49    | LPG 11.67kg cylinder             | Each       | Administered/utility | 3,921.730                     | 0.057   | 0.420                 | 0.427           |
| 50    | Telephone call charges           | Per Minute | Administered/utility | 1.790                         | 0.000   | 1.000                 | 1.000           |
| 51    | Toilet soap Lifebuoy (brand)     | Each       | Other non-food       | 117.292                       | 0.034   | 0.980                 | 0.991           |